

UNIVERSIDADE TÉCNICA DE LISBOA
INSTITUTO SUPERIOR TÉCNICO

UNIVERSITÉ CLAUDE BERNARD
LYON 1



Efficient algorithms for the identification of miRNA motifs in DNA sequences

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Dissertação para obtenção do Grau de Doutor em Engenharia Informática e de Computadores

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Junho de 2011

Acknowledgments

This thesis was the product of several years of work which, not unlike other academic endeavours, was punctuated by alternating periods of inspiration and frustration. It would not have been possible to complete this work without the support of several other people, at both the academic and personal levels.

This journey started at the Instituto Gulbenkian de Ciência, in the context of its pioneering PhD Program in Computational Biology. This program gave me the opportunity to contact many world-class experts in several topics of this field of research, paving the way to the choice of a project which would later result in this manuscript. In this regard, I have to thank the PhD Program and all the lecturers who took on to train a very heterogeneous group of young researchers which auspiciously included me. I must also thank the Fundação para a Ciência e a Tecnologia for funding my work by granting me a PhD scholarship.

I would also like to thank everyone at the BAOBAB Team of LBBE for kindly receiving me for the greater part of my doctoral work and the fellow researchers at KDBIO who accompanied me throughout these last few months and who created a truly welcoming environment.

I am in debt to several researchers who were kind enough to share their insights into my work over the years, namely Eric Westhof, Robert Giegerich, Eduardo Rocha, Rolf Backofen, just to name a few.

I am also deeply thankful to my family and friends for their relentless support, even in the face of my occasional negligence. And a special word to my mother, who always encouraged me and who has been nothing but thrilled that I chose the career she once dreamt of for herself.

I would also like to express my gratitude to the members of the jury who have been kind enough to accept being part of the thesis committee.

A final word of appreciation to my advisors, who always believed in my ability to complete this project in spite of my occasional hesitations. A special thanks to Ana Teresa Freitas, who has always nurtured my passion for research since the early days of my graduation thesis and who has always made me see the positive outlook of things. And to Marie-France Sagot, whose wisdom and uncompromising passion for science taught me, above all, that no matter how much a problem has been studied, there is always a fresh perspective and new insights to contribute with.

Abstract

Unravelling biological processes is dependent on the adequate modelling of regulatory mechanisms that determine the timing and spatial patterns of gene expression. In the last decade, a novel regulatory mechanism has been discovered and its biological importance has been increasingly recognised. This mechanism is mediated by RNA molecules named miRNAs that are the product of the maturation of non-coding gene transcripts and act post-transcriptionally usually to dampen or abolish the expression of protein-coding genes.

Despite having eluded detection for such a long time, it is now clear that the elucidation of the expression pattern of many genes cannot be achieved without incorporating the effects of miRNA-mediated regulation.

The technical difficulties that the experimental detection of these regulators entailed prompted the development of increasingly sophisticated computational approaches. Gene finding strategies originally developed for coding genes cannot be applied since these non-coding molecules are subject to very different sequence restraints and are too short to exhibit statistical properties that can be easily distinguished from the background. As a result, computational tools came to rely heavily on the identification of conserved sequences, distant homologs and machine learning techniques.

Recent developments in sequencing technology have overcome some of the limitations of earlier experimental approaches, but pose new computational challenges. At present, the identification of new miRNA genes is therefore the result of the use of several approaches, both computational and experimental.

In spite of the advancement that this research field has known in the last several years, we are still not able to formally and rigourously characterise miRNA genes in order to identify whichever sequence, structure or contextual requirements are needed to turn a DNA sequence into a functional miRNA.

Efforts using computational algorithms towards the enumeration of the full set of miRNAs of an organism have been limited by strong reliance on arguments of precursor conservation and feature similarity. However, miRNA precursors may arise anew or be lost across the evolutionary history of a species and a newly-sequenced genome may be evolutionarily too distant from other genomes for an adequate comparative analysis. In addition, the learning of intricate classification rules based purely on features shared by miRNA precursors that are currently known may reflect a perpetuating identification bias rather than a sound means to tell true miRNAs from other genomic stem-loops.

In this thesis, we present a strategy to sieve through the vast amount of stem-loops found in metazoan genomes in search of pre-miRNAs, significantly reducing the set of candidates while retaining most known miRNA precursors. Our approach relies on precursor properties derived from the current knowledge of miRNA biogenesis, analysis of the precursor structure and incorporation of information about the transcription potential of each candidate.

Our approach has been applied to the genomes of *Drosophila melanogaster* and *Anopheles gambiae*, which has allowed us to show that there is a strong bias amongst annotated pre-miRNAs towards robust stem-loops in these genomes and to propose a scoring scheme for precursor candidates which combines four robustness measures. Additionally, we have identified several known pre-miRNA homologs in the newly-sequenced *Anopheles darlingi* and shown that most are found amongst the top-scoring precursor candidates for that organism, with respect to the combined score. The structural analysis of our candidates and the identification of the region of the structural space where known precursors are usually found allowed us to eliminate several candidates, but also showed that there is a staggering number of genomic stem-loops which seem to fulfil the stability, robustness and structural requirements indicating that additional evidence is needed to identify functional precursors. To this effect, we have introduced different strategies to evaluate the transcription potential of the remaining candidates which vary according to the information which is available for the dataset under study.

Keywords: miRNA, gene finding, single-genome, robustness, stability, secondary structure

Resumo

A compreensão dos processos biológicos está dependente da modelação adequada dos mecanismos reguladores que determinam os padrões temporais e espaciais da expressão génica. Na última década, um novo mecanismo regulatório foi descoberto e a sua importância tem sido crescentemente reconhecida. Este mecanismo é mediado por moléculas de RNA designadas de miRNAs que são o produto da maturação de transcritos de genes não-codificantes e actuam de forma pós-transcricional de modo a, em geral, atenuar ou suprimir a expressão de genes codificantes.

Apesar da sua existência não ter sido detectada durante muito tempo, é hoje evidente que a descrição dos padrões de expressão de muitos genes não pode ser feita sem incorporar os efeitos da regulação mediada por miRNAs.

As dificuldades técnicas associadas à detecção experimental destes reguladores levaram ao desenvolvimento de abordagens computacionais cada vez mais sofisticadas. Estratégias de procura de genes originalmente desenvolvidas para genes codificantes não podem ser aplicadas porques estas moléculas não-codificantes estão sujeitas a constrangimentos ao nível da sequência de natureza muito diferente e são demasiado curtas para exibirem propriedades estatísticas que possam facilmente ser distinguidas das sequências circundantes. Daqui resulta que as ferramentas computationais tenham acabado por depender da identificação de sequências conservadas, de sequências homólogas distantes e do recurso a técnicas de aprendizagem.

Desenvolvimentos recentes no domínio da tecnologia de sequenciação superaram algumas das limitações das primeiras abordagens experimentais, mas introduzem novos desafios computacionais. Actualmente, a identificação de novos genes de miRNA é o resultado da aplicação de diversas abordagens tanto computacionais como experimentais.

Apesar dos avanços que este domínio de investigação conheceu nos últimos anos, ainda não é possível caracterizar genes de miRNA de um modo formal e rigoroso por forma a identificar os requisitos ao nível da sequência, da estrutura e do contexto genómico que são necessários para transformar uma sequência de DNA num miRNA funcional.

Os esforços computacionais desenvolvidos no sentido de enumerar o catálogo completo de miRNAs de um organismo têm sido limitados pela forte dependência que estes apresentam em relação a argumentos de conservação de precursores ou de semelhança de características. No entanto, os precursores de miRNA podem surgir ou desaparecer no decurso da história evolutiva de uma espécie e um genoma recém-sequenciado pode estar evolutivamente demasiado distante de outro genoma para se proceder a uma análise comparativa adequada. Por outro lado, a aprendizagem de regras de classificação complexas baseadas apenas em características partilhadas por precursores de miRNA que são actualmente conhecidos pode reflectir antes uma tendência para perpetuar um enviezamento quanto ao tipo de precursores identificados do que constituir efectivamente um forma de distinguir miRNAs de outras estruturas

semelhantes presentes no genoma.

Nesta tese é apresentada uma estratégia para filtrar o vasto número de potenciais candidatos identificáveis em genomas animais, reduzindo significativamente o seu número muito embora retendo a grande maioria dos precursores conhecidos. A abordagem usada apoia-se em propriedades exibidas pelos precursores e que foram deduzidas a partir do actual conhecimento do processo de biogénese dos miRNAs, na análise da estrutura secundária dos precursores e na incorporação de informação sobre o potencial de transcrição de cada candidato.

A abordagem foi aplicada aos genomas de *Drosophila melanogaster* e *Anopheles gambiae*, o que permitiu mostrar que há uma forte tendência da parte dos pre-miRNAs anotados no sentido de apresentarem estruturas secundárias robustas e propôr um esquema de classificação para candidatos que combina quatro medidas de robustez. Adicionalmente, foi possível identificar vários homólogos de pre-miRNAs conhecidos no genoma do recém-sequenciado *Anopheles darlingi* e mostrar que estes se encontram entre os candidatos melhor classificados naquele organismo em relação à combinação das medidas de robustez. A análise estrutural dos candidatos e a identificação da região do espaço das estruturas secundárias onde os precursores conhecidos se encontram permitiu a eliminação de muitos candidatos, mas permitiu também observar que um grande número de estruturas no genoma parece cumprir os requisitos de estabilidade, robustez e de estrutura secundária, ilustrando a necessidade de recorrer a meios adicionais para identificar precursores funcionais. Para este efeito, foram introduzidas diversas estratégias para avaliar o potencial de transcrição dos candidatos, que variam consoante a informação disponível para o genoma em estudo.

Palavras-chave: miRNA, procura de genes, genoma único, robustez, estabilidade, estrutura secundária

Résumé

La compréhension des processus biologiques est dépendente de la modélisation adéquate des mécanismes de régulation qui déterminent la répartition spatiale et temporelle de l'expression génique. Au début de la dernière décennie, un nouveau mécanisme de régulation a été découvert et son importance biologique a été de plus en plus reconnue. Ce mécanisme est médié par des molécules d'ARN appelés miRNAs qui sont le produit de la maturation de transcrits de gènes non-codants et qui agissent d'une façon post-transcriptionnelle afin d'atténuer ou supprimer l'expression de gènes codants.

Quoi qu'ils aient longtemps échappé à la détection il est maintenant évident que les variations de l'expression de nombreux gènes ne peuvent être comprises sans intégrer les effets de la régulation mediée par des miRNAs.

Les difficultés techniques de la détection expérimentale de ces régulateurs a suscité le développement d'approches computationnelles de plus en plus sophistiquées. Les stratégies initialement développées pour la recherche des gènes codants ne peuvent pas être appliquées car ces molécules non codantes sont soumises à très différentes restrictions au niveau de la séquence et sont trop courtes pour présenter des propriétés statistiques peuvant être facilement distinguées de celles des séquences parmis lesquelles elles se trouvent. Par conséquent, les outils de calcul comptent beaucoup sur l'identification des séquences conservées, sur les homologues lointains et les techniques d'apprentissage automatique.

Les développements récents dans la technologie de séquençage ont réussi à surmonter certaines des anciennes limites des approches expérimentales, mais posent de nouveaux défis informatiques. À l'heure actuelle, l'identification de nouveaux gènes de miRNA est donc le résultat de l'utilisation de plusieurs approches, à la fois computationelles et expérimentales.

Malgré l'avancement que ce domaine de recherche a connu dans les dernières années, nous ne sommes toujours pas capables de caractériser les gènes de miRNA de façon formelle et rigoureuse, de manière à identifier les contraintes au niveau de la séquence, de la structure ou du contexte génomique nécessaires à la formation de miRNAs fonctionnels.

L'utilisation d'algorithmes de calcul pour l'énumération de l'ensemble des miRNAs d'un organisme est limitée par une forte dépendance sur des arguments de conservation de précurseurs et de la similitude de caractéristiques. En effet, les précurseurs de miRNA peuvent apparaître *de novo* ou être perdus au long de l'histoire évolutive d'une espèce. De plus, un génome récemment séquencé peut être évolutivement trop éloigné d'autres génomes pour que l'on puisse faire une analyse comparative adéquate. Finalement, l'apprentissage des règles de classification complexes en se basant uniquement sur des caractéristiques communes aux plusieurs précurseurs de miRNA qui sont actuellement connus peut perpétuer un biais d'identification plutôt que de trouver vraiment un moyen de distinguer les miRNAs d'autres tiges-boucles génomiques.

Dans cette thèse, nous présentons une stratégie pour filtrer la grande quantité de tiges-

boucles se trouvant dans les génomes de métazoaires, réduisant de manière significative l'ensemble des candidats, tout en conservant la plupart des précurseurs de miRNA connus. Notre approche repose sur les propriétés des précurseurs provenant de l'état actuel des connaissances de la biogenèse des miRNAs, de l'analyse de la structure des précurseurs et de l'incorporation d'informations sur le potentiel de transcription de chaque candidat.

Notre approche a été appliquée aux génomes de *Drosophila melanogaster* et *Anopheles gambiae* ce qui nous a permis de montrer que dans ces génomes les pre-miRNAs annotés ont tendance à correspondre aux tiges-boucles robustes et aussi à proposer un systéme de notation des candidats qui combine quatre mesures de robustesse. Notamment, nous avons identifié plusieurs homologues de pre-miRNAs connus dans le génome récemment séquencé d'*Anopheles darlingi* et nous avons montré que la plupart se trouvent parmi les candidats à plus haute notation (selon notre combinaison de mesures) pour cet organisme. L'analyse structurelle de nos candidats et l'identification de la région de l'espace des structures où se trouvent généralement les précurseurs connus nous a permis d'éliminer plusieurs candidats, mais a également montré qu'il existe un nombre impressionnant de tiges-boucles génomiques qui semblent répondre aux exigences de stabilité, de robustesse et de structure. Ceci indique que des méthodes supplémentaires sont nécessaires pour identifier les précurseurs fonctionnels. Pour cela nous avons mis en place différentes stratégies pour évaluer le potentiel de transcription des candidats restants qui varient en fonction des informations qui sont disponibles pour les données analyseés.

Mots-clés : miRNA, recherche de gènes, génome isolé, robustesse, stabilité, structure secondaire

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Glossary

DNA	-	Deoxyribonucleic acid molecule composed of two strands of nucleotides forming a double helix. It is the carrier of genetic information necessary to all cellular activities.
RNA	-	Ribonucleic acid molecule.
Prokaryotes	-	Organisms which lack a nuclear envelope.
Eukaryotes	-	Organisms in possession of a nuclear envelope.
Chromatin	-	Complex of DNA and proteins found in eukaryotic cells.
Nucleus	-	Organelle found in eukaryotic cells which contains most of their genetic material.
Cytosol	-	Consists of the internal fluid of the cell where most of its metabolism occurs.
Aminoacid	-	Molecule that contains an amino and a carboxylic acid functional group.
Protein	-	Organic compound consisting of aminoacids joined by peptide bonds. It is essential to the structure and function of all living cells.
Enzyme	-	Biopolymer that catalyzes chemical reactions. Most enzymes are proteins although some are made of RNA or DNA.
Primary structure	-	the sequence of nucleotides of a nucleic acid molecule.
Secondary structure	-	the two-dimensional organisation of a nucleic acid molecule (usually RNA) specifying which nucleotides are forming base-pairs with one another. A common representation of secondary structure is the dot-parenthesis notation. Each unpaired nucleotide is represented by a dot '.', whereas a left-hand side of a base-pair is represented by a left parenthesis '(', and the right-hand side of a base-pair is represented by a matching right parenthesis ')'.
Stem-loop	-	also called hairpin, is a single-strand RNA structure folded on itself giving rise to a stem (a stack of base-pairs with possibly some intervening unpaired nucleotides called bulges or inner loops) and a terminal loop (a string of unpaired nucleotides at the middle portion of the sequence, flanked by both stem arms).